

Pablo Pena

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EDUCATION

University of Texas at Austin

B.S. in Mechanical Engineering Honors, Robotics Minor

August 2024 – May 2028

GPA: 4.0

SKILLS

CAD & Analysis: SolidWorks, Siemens NX, FEA, GD&T

Manufacturing: 3D Printing, Laser Cutting, SMT Soldering, Wire Harness Fabrication, CF Layups

Programming: Python, MATLAB, C#, Arduino, SQL, Java

PROFESSIONAL EXPERIENCE

BlackPearl Technology

Mechanical Engineering Intern

June 2026 - August 2026

- Reduced component count by 67% through design consolidation, streamlining manufacturability and reducing assembly complexity
- Reduced product assembly times by 60% through fixture and process improvements
- Designed and documented 30+ components and assemblies for production and testing applications using SolidWorks
- Managed configuration updates to a 100+ line-item BOM by coordinating engineering changes across multidisciplinary teams, including vendor updates, part revisions, standard hardware selection, and new component integration, ensuring accurate procurement and inventory documentation.

Landry's Inc.

Information Technology Intern

June 2024 - August 2024

- Developed a Power BI dashboard consolidating 9 key IT performance metrics, improving visibility and streamlining reporting across departments
- Reviewed and corrected 180+ intranet entries to improve data accuracy and maintain up-to-date operational information across restaurant locations
- Developed a C# application leveraging APIs to automate data transfers between loyalty systems, eliminating manual CSV processing

PROJECTS

RC Plane — Texas Design Build Fly

August 2024 - Present

- Designed and fabricated 4 fuselage prototypes using hybrid composite layups and vacuum bagging, optimizing stiffness-to-weight ratio and manufacturability
- Manufactured landing gear with carbon fiber/Kevlar with $\pm 45^\circ$ CF plies for increased torsional stiffness and 0° Kevlar plies for deflection resistance; validated layup quality through fit checks and assembly testing

Battle Bot

January 2025 - May 2025

- Designed and fabricated 10 custom components using SolidWorks, 3D printing, and laser cutting to create a sub-1 lb combat robot
- Selected and integrated drive, actuation, and power subsystems, including DC motors, servo actuation, compression springs, and a 7.4 V LiPo battery pack, to improve durability and weapon performance

ACHIEVEMENTS & AWARDS

- 6th Place Overall — AIAA Design/Build/Fly International Competition

April 2025

- Hispanic Scholarship Fund (HSF) Scholar

December 2024